Love of Variety and Dynamic Advertising

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Roadmap

1. Motivation
2. Model
3. Simulation Design
4. Results: Consumer Behavior
5. Next Steps

Motivation: Data Privacy, Ad Targeting, and Consumer Agency

- Amazon anti-trust scrutiny
  - Amazon accounts for almost 40% of US e-commerce sales (Statista, 2023)
  - Able to use consumer data to target ads and build market share as both retailer and platform
- Consumer agency over data (opt out)
  - Apple iOS 14.5 update: 85% opt-out rate upon rollout (Flurry Analytics, 2022)
  - Facebook ad revenue down $13bn in 2022 due to app tracking transparency (ATT) policy

Key Concern

How do data privacy policies that limit firms' ability to target ads based on consumer data impact consumer welfare and firms' incentives to invest in product quality and advertising?

Why Does Variety Matter?

Research Question

How do consumer preferences for variety impact the firms' targeting strategy and what are the welfare implications of data privacy policies that limit ad targeting?

- Understanding the role of variety-seeking is valuable for modeling consumer behavior and evaluating the impact data privacy policies on both consumers and firms
- Canonical logit models ignore consumer preferences for variety, which are empirically important
- These preferences may shape consumer switching behavior, which in turn shapes firms' incentives to strategically target ads and promotions (i.e. to encourage switch to a higher markup product)

What This Paper Does (So Far)

1. **Model:** Dynamic discrete choice model of consumer behavior with love of variety and strategic advertising by firms
2. **Simulation:** Characterize consumer and firm behavior across preference regimes and advertising specifications
3. **Estimation (coming):** Bring model to Nielsen scanner data to recover structural parameters
4. **Counterfactuals (coming):** Study advertising targeting policy and welfare implications

Today's talk: Consumer demand function + simulation results

Literature Connections

Demand side

- Lattin (1987): State dependence in consumer choice
- Chintagunta (1999): Extension to panel data
- Dubé, Hitsch, Rossi (2010): Habit formation and switching costs

Advertising / firm side

- Shaffer and Zhang (1995): Theoretical basis for ad targeting in a two-period model
- Erdem & Keane (1996): Learning dynamics
- Zhang, Choi, Cheng (2025): Updated targeting game

Contribution

Unite the demand and supply side literatures by modeling consumer preferences for variety in a dynamic discrete choice framework with strategic advertising.

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Consumer's Problem

Consumer i chooses product j in period t to maximize:

$$U_{ijt} = \underbrace{\beta_{ij} X_{jt}}_{\text{characteristic component}} + \underbrace{\gamma_i \log(1 + \Sigma_{ijt}^2)}_{\text{novelty component}} - \underbrace{\alpha\, p_{jt}(1 - a_{jt})}_{\text{ad-adjusted price}} + \xi_j + \varepsilon_{ijt}$$

- where $\Sigma_{ijt} = X_{jt} - \theta_{it}$ is the deviation of product j's characteristic from consumer i's previously chosen characteristics θ_{it} in period t.
- at the moment, $\theta_{it} = \frac{1}{t} \sum_{s=1}^{t-1} X_{js}$, or the mean of previously chosen product characteristics.

Key parameters:

- $\beta_i > 0$: quality
- $\gamma_i > 0$: variety-seeking

Choice probability (MNL):

$$P_{ijt} = \frac{e^{U_{ijt}}}{1 + \sum_k e^{U_{ikt}}}$$

Follows from $\varepsilon_{ijt} \sim$ Gumbel$(0, 1)$

State Variable: History of Choices

History of Choices θ_{it}

$$\theta_{it} = f(\{X_{js}\}_{s<t}) = \frac{1}{t}\sum_{s=1}^{t-1} X_{js}$$

The consumer's history is summarized by the mean of previously chosen product characteristics.

Simulation design:

1. **Initial state:** Simulate $T_{\text{prior}} = 5$ pre-sample periods, choose products from $J \in [1, 2, 3, 4, 5]$ to generate an experience-based prior.

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Simulation Parameters

Table: Parameters

Param.	Value	Description
T	100	Periods
J	5	Products
S	1,000	Simulations
β_{ij}	2	Quality
γ_i	$\{6, 9, 12\}$	Variety-seeking
α	0	Price (off)
ξ_j	0	Unobserved quality (off)
β_{df}	0	Discount factor (off)

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Utility Measurement (No LOV)

Utility Function without LOV

$$\hat{P}_{jt} = \beta_{ij} X_{jt} + \epsilon_{ijt}$$

- Any variation due to ε_{ijt}, not consumer preferences
- CCPs are not informative about consumer behavior or firm incentives
- Nearly all mass on Product 5, which has the highest quality X_{jt}

Table: CCPs (No LOV)

Product	CCP
Product 1	0.02%
Product 2	0.54%
Product 3	4.05%
Product 4	22.3%
Product 5	72.0%

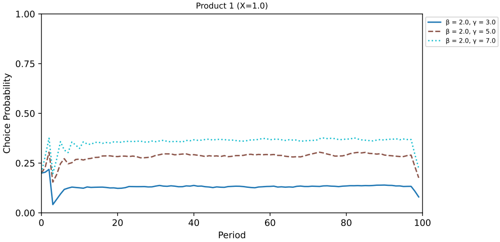
Consumer Behavior with LOV

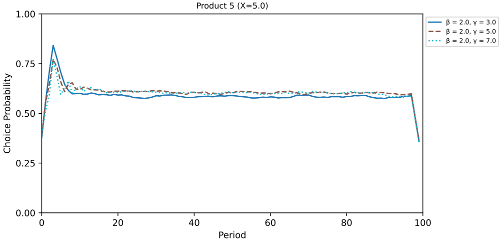
Utility Function with LOV

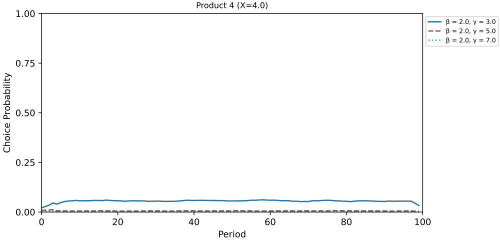
$$\hat{P}_{jt} = \beta_{ij} X_{jt} + \gamma_i \log(1 + \Sigma_{ijt}^2) + \epsilon_{ijt}$$

- $\Sigma_{ijt} = X_{jt} - \theta_{it}$ captures the deviation of product j's characteristic from consumer i's previously chosen characteristics θ_{it} in period t.
- Consumer behavior varies significantly across variety specifications
- Higher γ_i leads to more switching
- Cross-sectional sorting: consumers with higher γ_i are more likely to switch to lower indexed products

LOV Specification







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Road to Estimation

1. **Multiple consumers**
   Extend simulation to $I = 5$ consumers who have heterogeneous (β_i, γ_i); aggregate to market shares
2. **Consumer value function**
   Move from reduced-form simulation to solving the full dynamic discrete choice with continuation values
3. **Nielsen scanner data**
   Match simulated moments to data; identify (β_i, γ_i) using BLP instruments
4. **Firm's Problem**
   Ad targeting policy; profit maximization; counterfactuals

Summary

1. **Model:** Consumer utility depends on deviation from experiences; nests variety-seeking
2. **History:** Rich behavioral dynamics; regime differences generate cross-sectional sorting
3. **Next:** Multiple consumers, structural estimation, Nielsen data

Thank you — questions welcome

Appendix

References

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Simulation Algorithm (Consumer)

1. Draw $\varepsilon_{ijt} \sim \text{Gumbel}(0, 1)$ for all i, j, t
2. Initialize $\bar{X}_{i0}$ (set mean or prior mean)
3. For $t = 1, \ldots, T$:
    1. Compute $U_{ijt} = \beta_i X_{jt} - \gamma_i (X_{jt} - \bar{X}_{it})^2 + \kappa a_{jt} + \varepsilon_{ijt}$
    2. Set $j^* = \arg\max_j U_{ijt}$
    3. Update $\bar{X}_{it+1} = \frac{1}{t} \sum_{k \leq t} x_{ijk^*}$
4. Compute CCPs via log-sum-exp for numerical stability:
   $\log P_{ijt} = U_{ijt} - \log \sum_k e^{U_{ikt}}$
5. Average over $S = 1{,}000$ simulation draws